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Multiclass Multiple-Instance Learning for Predicting Precursors to Aviation Safety Events

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In recent years, there has been a rapid growth in applying machine learning techniques that leverage aviation data collected from commercial airline operations to improve safety. Anomaly detection and predictive maintenance have been the main targets for machine learning applications. However, this paper focuses on the identification of precursors, which is a relatively newer application. Precursors are events correlated with adverse events that happen before the adverse event itself. Therefore, precursor mining provides many benefits, including the identification of factors relevant to the occurrence of an adverse event and their signatures, which can be tracked throughout a flight to alert the operators of the potential for an adverse event in the future. This work proposes using the multiple-instance learning framework, a weakly supervised learning task, combined with a carefully designed binary classifier leveraging a Multi-Head Convolutional Neural Network–Recurrent Neural Network (MHCNN-RNN) architecture. Multiclass classifiers are then created and compared, enabling the prediction of different adverse events for any given flight by combining binary classifiers, and by modifying the MHCNN-RNN to handle multiple outputs. Results obtained showed that the multiple binary classifiers perform better and are able to accurately forecast high-speed and high-path-angle events during the approach phase. Multiple binary classifiers are also capable of determining the aircraft parameters that are correlated to these events. The identified parameters can be considered precursors to the events and may be studied/tracked further to prevent these events in the future.

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I. Introduction

THE aviation industry brings tremendous social and economic benefits. It has been observed that the size of the air transportation industry has doubled every 15 years [1] and was expected to continue growing [2] before the COVID-19 pandemic. In 2018 alone, airlines around the world carried a total of 4.3 billion passengers, whereas the total global economic impact of the industry reached USD 2.7 trillion in 2016 [1]. Furthermore, even as the industry continues to grow and more people are flying every day, aviation safety has improved over the past decades, as seen in Fig. 1. The number of fatalities for commercial air carriers decreased from 81 per 100 million persons on board in 1996 to 0.6 in 2019, which is below the target rate set by the Federal Aviation Administration (FAA).

To significantly improve safety, the aviation industry has recently been moving toward a proactive approach to safety assessment and vulnerability identification, which includes characterizing potential risks in terms of anomalies or deviations from nominal operations and the precursors to adverse events. This knowledge can be leveraged to increase awareness of emerging vulnerabilities among the operators and be incorporated into automated monitoring tools to flag the risks before they result in near-misses, incidents, or accidents [4].

Extensive data collection and advancement in data-mining methodologies are key enablers to proactive risk management in aviation [4,5]. Airline programs such as the Flight Operations Quality Assurance (FOQA)¶ have enabled the creation of large and heterogeneous data sets. FOQA is a voluntary safety program that is designed to make commercial aviation safer. Traditional techniques of flight data analysis have focused on a continuous cycle of data collection from on-board recorders, retrospective analysis of flight-data records, identification of operational safety exceedances, design and implementation of corrective measures, and monitoring to assess their effectiveness. Airlines work with the FAA to reduce and eliminate safety risks, and flight safety divisions within airlines generally use FOQA data to perform exceedance and statistical analyses [6].

Research efforts have been made toward advancing the development of data-driven methodologies, applying data science techniques, and using modern machine learning, including deep learning algorithms, in the context of aviation safety [7–15]. The majority of the data mining effort in aviation is directed at detecting anomalies in aviation data [8,9,16–19] and leveraging unsupervised learning techniques due to a lack of labeled data. Data mining has also been applied toward predictive maintenance in aviation [7] and predicting future trajectory states [20]. While identifying anomalies is essential, it is also critical to investigate the causal factors or precursors to these anomalies or other safety events to better understand them and prevent them in the future. Recent literature has focused on identifying precursors to safety events and anomalies using FOQA data [21–23]. Precursors can be defined as any events correlated to a safety incident and occur prior the incident itself [21]. They help forecast safety events and provide insights into why the event happened. Knowledge of the precursors can then be used to initiate actions to avoid events from occurring [23]. Therefore, identifying and monitoring precursors is an essential step toward proactive safety enhancement in aviation operations. One of the main challenges in identifying precursors observed in the literature is the lack of subject-matter-expert-validated labels for the identified anomalies and safety events. To account for the lack of labels usually encountered, this paper proposes applying a weakly supervised

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¶FOQA Program URL: https://www.faa.gov/documentLibrary/media/Advisory_Circular/AC_120-82.pdf.

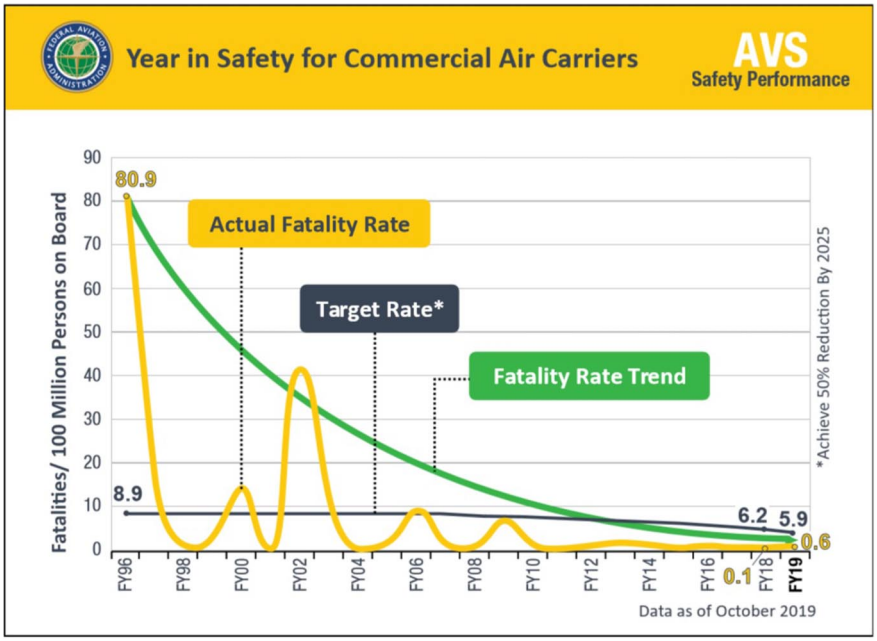

Fig. 1 Commercial aviation fatal accident rate from FY96 to FY19 [3].

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learning technique called multiclass multiple-instance learning (MIL) [24] to detect different multiple adverse events and their precursors.

This paper aims to develop a methodology that leverages highly dimensional aviation data to predict multiple adverse events and identify their precursors. This work will use a flight-level label called a bag label in MIL terminology to predict the anomalies at the flight level and identify their precursors and their occurrence in time during a flight. The current framework uses a deep learning model constructed as a multiheaded convolutional neural network (CNN) where each flight sensor gets its own CNN. The multiheaded architecture is designed to work with the MIL framework. Successful implementation of the methodology uses the flight's data to 1) predict adverse events and 2) determine the precursors to the predicted adverse events. The first major benefit from this framework over current approaches will be the extension of the binary classifier to perform multiclass predictions. The second benefit is the ability to retrieve the precursors with little to no postprocessing due to better transparency and explainability (compared with previous architectures), which could help analysts find root causes of safety events faster.

II. Background

This section presents recent work conducted related to anomaly detection, precursor mining, and other applications of machine learning and deep learning models in aviation safety. The data source used for developing the model presented in this work is also introduced, and lastly, the definition of multiple-instance learning and its accompanying assumptions.

A. Previous Work

Focusing on precursors to anomalies is important as it allows identifying potential causes of safety hazards. Recent work shows a growing interest in detecting precursors in various domains. Multiple techniques have been explored, and the most relevant ones are summarized in this subsection.

Ning et al. [25] presented an approach for precursor identification using nested multi-instance learning (n-MIL). In their work, they forecast societal events in different cities. At the instance level, the probability of an article published on a given day is modeled using a simple logistic function. These probabilities are then aggregated over a day. Finally, the probabilities for different days are aggregated together up to a certain number of days before the event, creating the nested structure. The authors explain that the probability of a news article on a given day can be used to estimate how related an article is to the target. Because the articles with higher probabilities influenced the classifier decision, they are likely to be precursors of the predicted societal event.

Janakiraman [21] proposed to use a deep temporal multiple-instance learning (DT-MIL) framework, which combines multiple-instance learning and recurrent neural networks (RNNs), in this case a gated recurrent unit (GRU), to mine precursors in FOQA data. In this approach, the individual time steps are considered low-level instances, and the whole flight is considered a bag. The labels (occurrence of an adverse event) are given at the bag level. However, the methodology takes advantage of MIL and uses the low-level instances to correctly predict the bag label, allowing to infer the instance-level label. The time steps at which the probability of a safety event occurring is greater than a defined threshold are retrieved. The region of time for which the probability (called the precursor score) of a safety event occurring is high is then analyzed during postprocessing, where each feature is perturbed one at a time. The precursors are identified by finding the features whose perturbations had more significant impacts in reducing the precursor score. The DT-MIL model is also referred to as a newer version of the Automatic Discovery of Precursors in Time Series (ADOPT),** which is different from the architecture presented in [26].

Ackley et al. [22] used a sequential backward selection technique and random forest classification models to predict unstable approach adverse events. They identified the critical parameters using a cumulative feature importance score and grouped them into various categories of parameters (such as energy-related, configuration-related, and more) that contribute significantly to identifying the adverse event. Their analysis is conducted at fixed altitudes above the event detection trigger altitude of 1000 feet above touchdown. Similarly, Lee et al. [27] used a random forest algorithm to identify precursors to two different aviation safety events using supervised learning. The normalized precursor score is obtained using a Gini importance for all parameters contained in the classification model.

Melnyk et al. [28] proposed a framework for detecting precursors to aviation safety incidents due to human factors based on hidden semi-Markov models. They performed an empirical evaluation of their models against traditional anomaly detection algorithms and demonstrated better performance on synthetic and flight simulator data.

**https://github.com/nasa/ADOPT.

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Table 1 Adverse events labeling

Adverse event	Comment
High speed in approach	Flagged at 1000 ft
Low speed in approach	Flagged at 1000 ft
High rate of descent in approach	Flagged between 1000 and 500 ft
High bank in approach	Flagged between 1000 and 400 ft
High path angle in approach	Flagged at 1000 ft
Low path angle in approach	Flagged at 1000 ft
Deviation from localizer	Flagged between 1000 and 500 ft
Deviation below glideslope	Flagged between 1000 and 500 ft
Flaps late setting at landing	None

B. DASHlink Flight Data

The proposed framework is demonstrated using a publicly available data set obtained from NASA's DASHlink website, a collaborative sharing network for researchers in the data mining and systems health management field.[††] Flight data were recorded from a single type of regional jet operating in commercial service over three years. The data contain detailed aircraft flight condition data, system performance, and other engineering parameters but are de-identified such that it cannot be traced back to a particular manufacturer or airline. The parameters that are available in the DASHlink data set fall into six categories: atmospheric data (pressure altitude, air temperature, wind speed, etc.), attitude data and corresponding rates (angle of attack; bank, pitch, and drift angles; glideslope deviation; etc.), GPS data (longitude and latitude, altitude, altitude rate, etc.), engine data (exhaust gas temperature, oil temperature, fuel flow. etc.), control data (air brake, aileron, elevator, flap positions, etc.), and navigation data (selected airspeed, selected heading, selected altitude, N1 target, etc.). Because this data set is not part of any airline's FOQA program, additional preprocessing is required to create FOQA-like flags and label safety events for individual flights, where the labeling was created by using domain-based rules. The definitions of resulting adverse events are presented in Table 1.

Each of these events is characterized by a severity level ranging from 1 to 3, with three being the most severe. For this work, only flights without any safety events and the ones with safety events with a severity level 3 were considered. This filtering allowed for limited overlaps between normal and abnormal operations. Moreover, to ensure a multiclass problem instead of a multilabel one, flights with multiple adverse events were omitted. When downloading the DASHlink metadata, a total of 141,585 flights are available. The high-speed and high-path events accounted for most of the flights that experienced an event of severity level 3. These flights were included in the data set used to train, validate, and test the model developed in this paper. Because the most significant portion of the DASHlink data corresponded to flights with no anomalies (62% of the total number of flights), a subset of nominally operated flights matching the number of high-speed events was included in the final data set used. The discrepancy observed in Fig. 2 between the number of nominal and high-speed flights is due to failures during the flights' download. It is noted that the proportion of adverse outcomes is significantly small in reality, and a high proportion of adverse events in the data set could point to the thresholds set for identifying them to be too tight. Therefore, this paper focuses on the methodological aspect of using the data and events. Finally, it is noted here that adverse events may not necessarily lead to adverse outcomes (i.e., accidents) but rather serve as an early warning system to undertake corrective measures.

C. Multiple-Instance Learning

The multiple-instance learning framework is used when labels are only available in sets called bags [29], and each bag contains many instances. Therefore, the learning task is supervised, but because the labels are not provided for each instance, we say that the task is weakly supervised. This framework is of interest because it alleviates the burden of requiring labels. In particular, it is useful in the context of aviation because of the lack of labeled data. In this context, we can consider the bag to be a flight and the instances to be the time steps of the features.

Furthermore, positive bags are flights with a safety event, whereas negative bags have no events recorded. The standard MIL assumption, which is used in this work, states that all negative bags contain only negative instances and that all positive bags contain at least one positive instance [29]. That is to say that a flight that experiences an event has at least one abnormal time step that caused the whole flight to be considered a positive bag. Flights can later be divided into training, validation, and testing sets, and the performances of a chosen model can then be evaluated on those two subsets.

The learning task for this work is a classification problem. In a classification application, the task is to use given data and assign it to multiple predefined classes. In particular, given the time series from sensor data, we can assign a label such as a safety event [8]. MIL classification tasks can be performed at the bag and instance level. However, the methodology developed in this work will use the bag label to infer the instance label of each feature at each time step. In other words, for example, if we classify a bag as positive, the algorithm will determine (infer) the time instances that enabled the classification of the bag.

III. Methodology

The intelligent methodology for the discovery of precursors of adverse events (IM-DoPE) developed in this work is described in this section. The four steps of the methodology, as seen in Fig. 3, are the data processing, the development of the model, the extraction of precursors, and precursor scores' derivation. These steps allow for the discovery of precursors related to adverse events of interest, which can be used to provide insights into the cause of these events.

A. Data Processing

The DASHlink sample flight data are recorded at 1 Hz frequency; therefore, no effort was placed in resampling the data set at the same frequency, as this is a sufficiently high frequency. The data set had already been cleaned before accessing it, as no missing data was observed. Subsequent tasks mainly focused on selecting events of interest, dimensionality reduction, data interpolation, and data formatting.

1. *Event Definition and Selection*

As previously stated, two adverse events were selected for this work due to their higher frequency in the available data set. These events are predefined and characterized by the deviation of specific aircraft parameters from accepted nominal behaviors. Figure 2 shows the counts of nominal flights and adverse flights that experienced one of the two events. As seen in the figure, the high-speed event was the most commonly observed event among the available data, followed by the high-path-angle event.

2. *Feature Selection*

The dimensionality of the data was reduced by leveraging the correlations among the features. Certain parameters (e.g., computed airspeed and true airspeed) are different but convey similar information. This is due to the redundancy of parameters, the aircraft's physics, and the derivation of parameters [22]. A correlation matrix was created using the data. For each pair of features, the Pearson's correlation coefficient is computed. The coefficient is a measure of the strength of the association between the features in the pair.[‡‡] The Pearson correlation coefficient for two random variables X and Y is given by

$$\rho = \frac{\sigma_{XY}^2}{\sqrt{\sigma_{XX}^2 \sigma_{YY}^2}} \tag{1}$$

[††]https://c3.nasa.gov/dashlink/resources/?page=3&sort=-created&type=28.

[‡‡]Data Analysis—Pearson's Correlation Coefficient: http://learntech.uwe.ac.uk/da/Default.aspx?pageid=1442.

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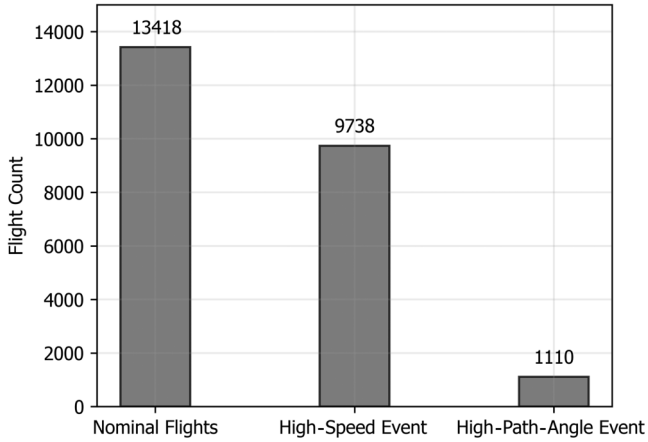

Fig. 2 Flight counts for nominal and adverse flights.

The correlation coefficient ρ is bounded between $-1 < \rho < 1$, and values closer to 1 signify a strong positive correlation, whereas values closer to -1 suggest a strong negative correlation. In Eq. (1), σ^2_{XX} is the sample variance for the variable X, and σ^2_{YY} the one for the variable Y, and σ^2_{XY} the covariance between X and Y. Because features are considered highly correlated if they have a correlation coefficient higher than 0.9 [30], a threshold of $\rho = 0.90$ was set so that for a given pair of features, with the absolute value of their correlation coefficient greater or equal to the threshold, one of the correlated features was removed. Note that the absolute value of the correlated coefficient between feature pairs was taken because features can be highly negatively correlated. For this work, only nonhighly correlated continuous variables were used as the input feature space. In addition to the correlation-based feature selection, trivial precursors were removed. This step was performed to avoid observing one of the speed parameters as precursors to the high-speed event.

3. *Data Resampling*

Resampling was performed to convert the data at 0.25 n mile distance intervals. Uniform distance intervals are favored since pilots prefer to think in terms of distances because many procedures are defined based on distance, in particular holding procedures [31]. Additionally, resampling enables a consistent number of time steps from one flight to another. In fact, for the same distance (e.g., 20 n miles), faster flights generate fewer data points than slower flights. Consistent time steps are preferred because they allow for easier processing of the data by deep learning frameworks such as Pytorch or TensorFlow. Thus, starting from a distance of 20 n miles before crossing 1000 ft above the landing surface, flights were resampled at every 0.25 n mile. Each flight, therefore, contains 81 data points or time steps. A Python code was developed to interpolate each feature to retrieve their values at every 0.25 n mile, yielding a data set similar to Table 2 for each flight. Figure 4 shows two examples of interpolated data. The original flight data contained 281 points for each feature, whereas the interpolated data contain only 30 points.

4. *Data Formatting*

In practice, modern algorithms such as CNNs and RNNs usually take as inputs three-dimensional tensors. To account for this, data from each flight, as specified in Table 2, are concatenated together into a tensor as shown in Fig. 5, with $X^f_{i,j}$ referring to feature j for flight f at time step i.

The data take a three-dimensional form where the first dimension N corresponds to the batch size or the number of flights that the deep learning algorithm will process at once. The second dimension L is the number of time steps in each flight (81 for this work). Finally, the last dimension D refers to the number of features/aircraft parameters corresponding to 58 continuous variables.

B. Model Architecture

The model architecture was chosen to take advantage of the feature extraction capabilities of CNNs [32,33] and the temporal pattern recognition ability of RNNs, yielding an MHCNN-RNN architecture. This architecture choice allows the aircraft's parameters to be

Table 2 Example results of interpolating parameters for a given flight

Distance away from 1000 ft	Feature 1	Feature 2	$\cdots$	Feature d
20	X	X	$\cdots$	X
19.75	X	X	$\cdots$	X
$\vdots$	X	X	$\cdots$	X
0	X	X	$\cdots$	X

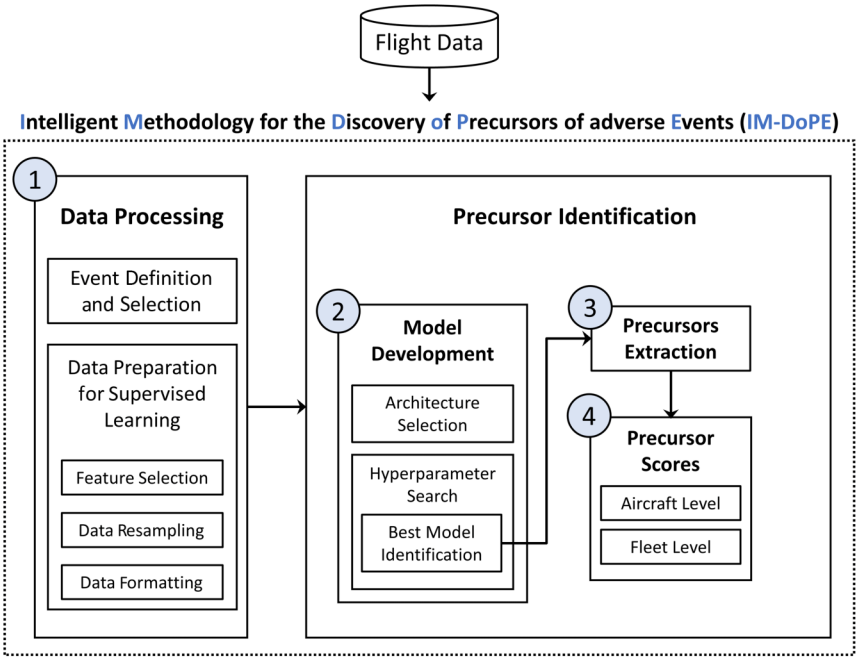

Fig. 3 Intelligent methodology for the discovery of precursors of adverse events.

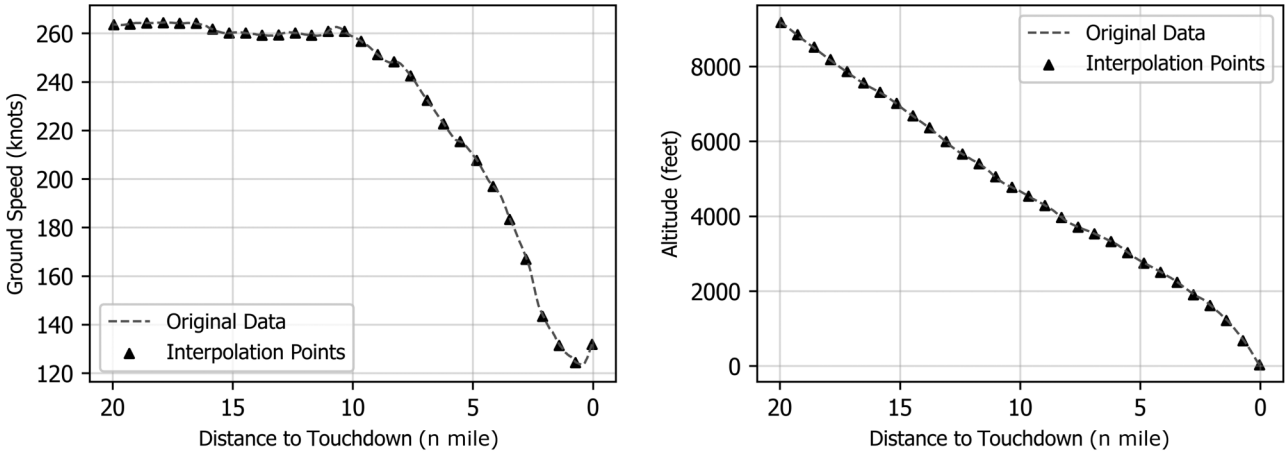

Fig. 4 Example of altitude and ground speed interpolations.

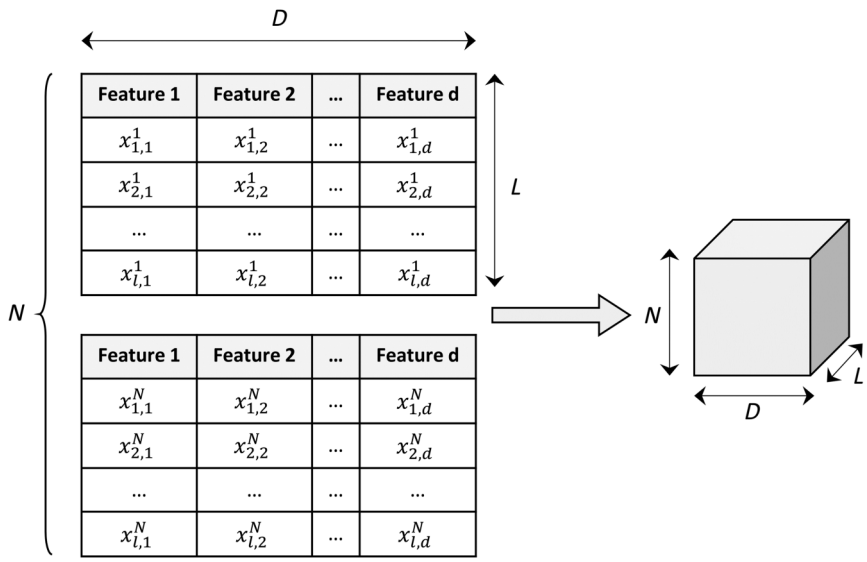

Fig. 5 Flight data reshaping.

initially processed individually by the MHCNNs, which yields feature maps. The RNN then learns the complex (time-dependent) correlations between the feature maps. As explained in [32], the MHCNN-RNN structure has multiple benefits, such as enabling each head to be adjusted to each sensor data and allowing for a flexible architecture robust to changes (add, replace, remove) in sensor configurations. However, the main advantage of the architecture for this paper is the enhanced feature extraction capability for each sensor, enabling the recognition by the CNN of each sensor's importance toward the final model prediction. The model was developed using PyTorch [34], a high-performance Python deep learning library.

1. *Architecture Selection*

The MHCNN-RNN architecture combines a Multi-Head CNN (MHCNN) to a GRU, a type of RNN, as shown in Fig. 6.

CNNs are a specialized type of neural network that can be used to process data with grid-like topology, where time series can be thought of as 1-D grids. The CNNs were chosen because they are known to extract meaningful features from data using their convolutional layers. In particular, for time series, the convolutional layers do so by sliding a trainable kernel (weights) across the time series and generating feature maps [35]. The operations performed are element-wise dot products across multiple windows of the time series. Hyperparameters such as the kernel size (known as the dimension of the sliding window) and the stride (which indicates how many time steps the kernel skips before performing a new operation) have an important effect on the CNN capacity to retrieve and extract features. Another important hyperparameter is the number of channels. In practice, univariate time series can be thought of as having only one channel, whereas multidimensional time series have multiple channels. Because a multihead structure is chosen, the number of channels for each aircraft parameter is changed through the convolution operation. Though these channels have no interpretable meanings, they represent some aspect of the time series information, allowing for better feature extraction.

Four CNNs are used in each head to process the time series and expand the unique channel of each time series, as seen in Figs. 6 and 7, resulting in the extraction of relevant information from each window of the time series. On the one hand, in Fig. 6, k_1, k_2, and k_3 represent the number of channels. On the other hand, W_{N_1}, W_{N_2}, and, W_{N_3} represent the time series length after each convolution operation. After each operation, the length of the time series depends on CNN parameters such as the kernel size, the stride, and whether or not padding was used. The first three convolutional layers use batch normalization to reduce the internal covariance shift. A byproduct of this shift is the added regularization effect [32]. Each normalization is followed by a ReLU activation function, which has become a default function when developing neural networks due to its many advantages [36]. The fourth convolutional layer reduces the number of channels back to 1. It applies a sigmoid activation function, bounding the output (feature map) between 0 and 1 and allowing it to be interpreted as the probability of a given feature being a precursor. The chronological order of the feature maps was kept, which

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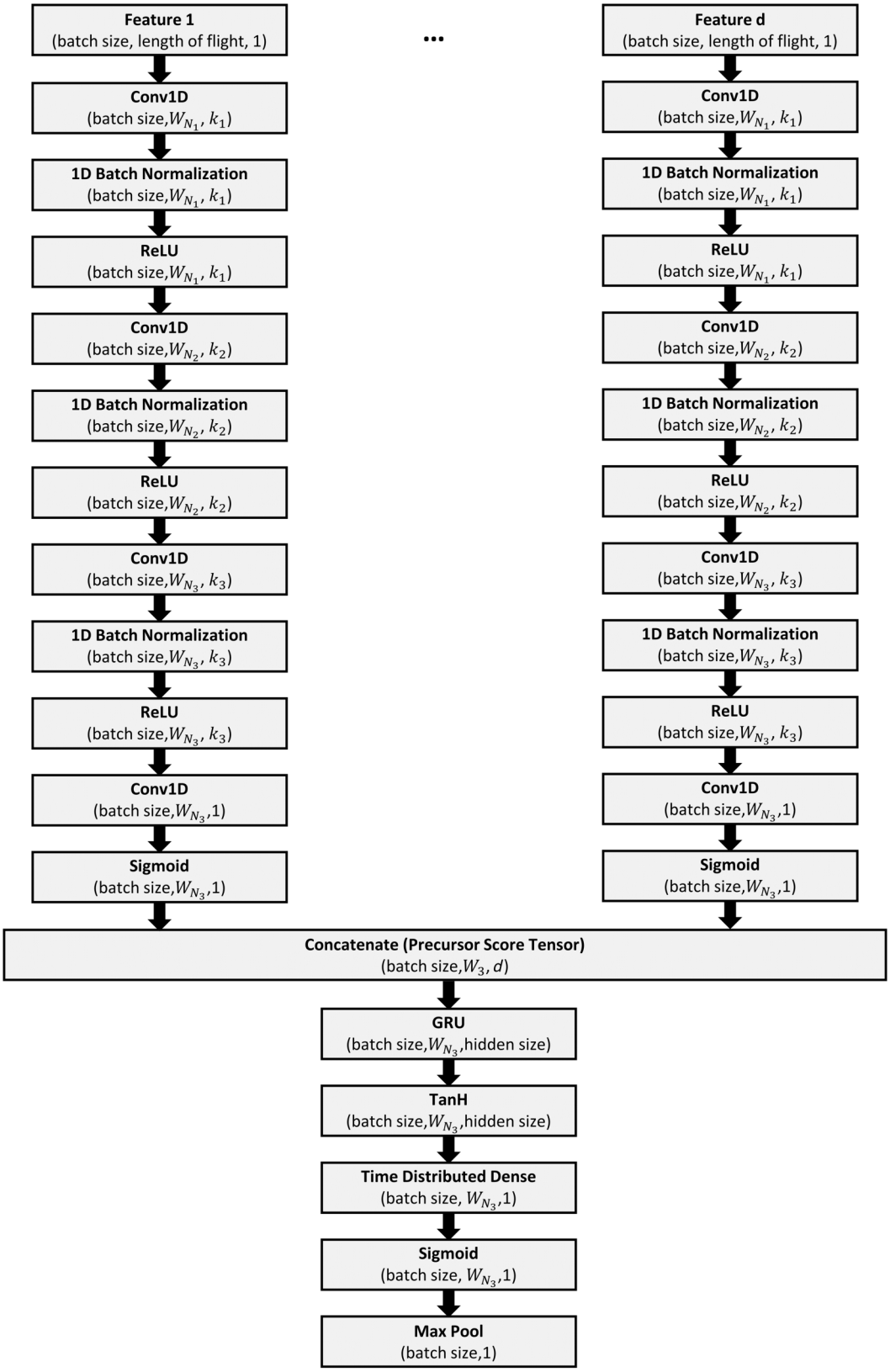

Fig. 6 Layer configuration with output size of layer.

helped maintain the temporal information of the data. A concatenation step was then performed to combine the feature maps of each aircraft parameter into a three-dimensional precursor score tensor.

The concatenated tensor is the input to the GRU. Unlike the MHCNN that processes the features independently, the GRU processes the features altogether. This layer learns the temporal patterns present in the extracted information by the previous convolutional layers. The GRU is appropriate for handling temporal data due to its structure characterized by two gates: the update and the reset gates [33]. These gates are used to select which information to keep or ignore and allow the GRU to handle long-term dependencies better than simpler architectures [33]. A hyperbolic tangent activation is then applied to the output of the GRU and followed by a time-distributed dense layer that squeezes all the features into one unique

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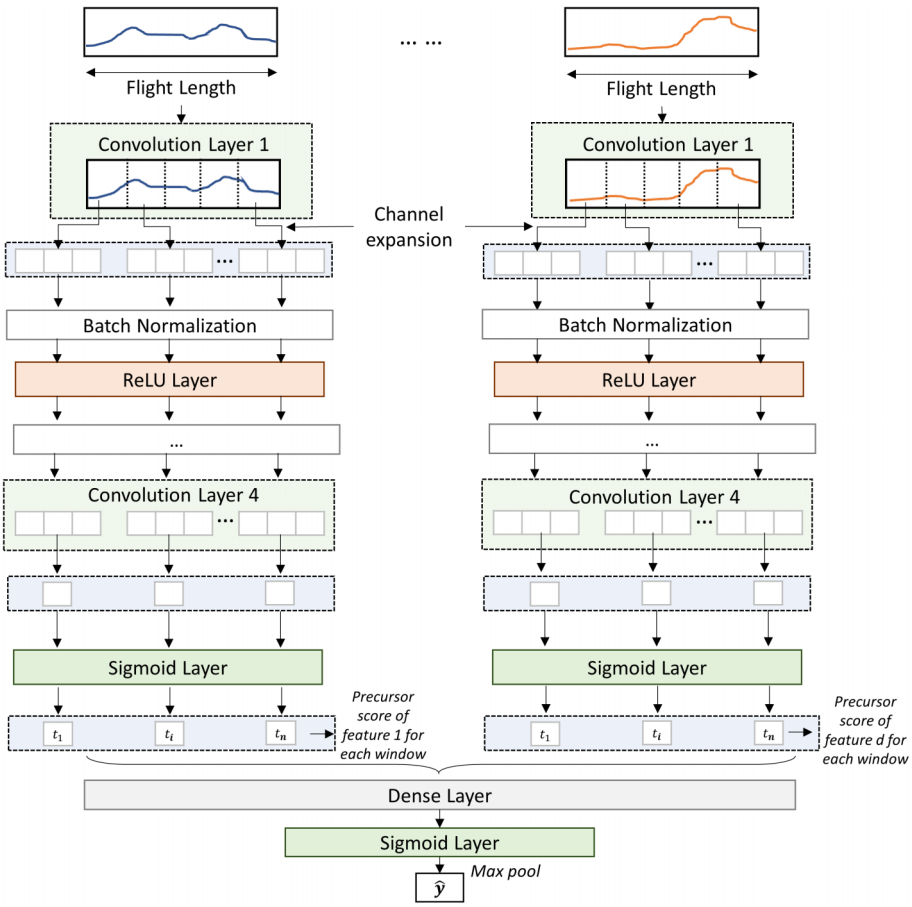

Fig. 7 Layer configuration showing where the precursor score is extracted.

dimension, allowing for additional approximation capability [21]. The output is passed through a sigmoid function and can be interpreted as the precursor probability over time. Finally, to classify a given flight as positive (the event occurred), it is assumed that the whole flight would be labeled as positive if any instance of time is positive (MIL assumption). Thus, a max-pooling layer is used to get the maximum probability across time to classify the flight. A threshold is set such that a flight is positive only if the maximum probability across time is greater or equal to the threshold. The architecture is thus expected to achieve two critical goals:

1) Label the bag for identifying adverse events.

2) Extract precursors and the time instance at which they occur.

Although the two approaches are similar because they take advantage of the MIL framework for precursor mining, it is essential to note the differences between this architecture and the previously developed algorithm ADOPT [21]. The architecture proposed in this paper is designed such that the precursor probabilities for each feature can be extracted directly from the neural network layers. This approach aims at creating a more interpretable model. ADOPT can only identify temporal precursors and is required to perform a sensitivity analysis during a postprocessing step, which consists of perturbing one unique feature at a time and measuring its effect on the time instance probabilities to determine precursors in the feature space. This method potentially misses interactions between features. Thus, retrieving precursors by leveraging the combination of the MHCNNs (extraction of individual information for each parameter) and the GRU (extraction of temporal interactions of parameters) mitigates this drawback while eliminating the overhead required to set up the postprocessing analysis.

2. *Multiclass Extension*

a. Multiple Binary Classifiers. The architecture described in III.B.1 corresponds to a binary classifier. Both IM-DoPE and ADOPT use binary classifiers as their task is to predict the occurrence of an adverse event correctly. Both models could be extended to include multiple outputs allowing for the training of multiple events at once [21]. However, the multiclass model is obtained for this work by combining multiple binary classifiers because they performed well at completing their individual tasks. Two popular approaches of combining the classifiers are the "one-vs-all" and the "one-vs-one" strategies [37]. The first strategy learns to classify one class when the class is distinguished from the other ones. The second strategy requires creating multiple classifiers for different pairs of classes. In this work, the approach taken is a mix between the two strategies. Indeed, two classifiers are trained for each event, similar to the "one-vs-all" approach. However, each of them is trained to either recognize the event or the default nominal operation. This approach was chosen to allow for the interpretation of nominal flights when performing predictions and to avoid training an extensive number of models. When using the models to perform an inference, the output class is chosen to be the output of the model with the highest probability (i.e., the highest confidence) that is greater than a specified threshold. Suppose that none of the probabilities are greater than the threshold. In that case, the output is defaulted to the class corresponding to normal operations because each model either predicts an event or normal operation. When using sigmoid functions, the decision threshold is problem dependent and could be treated as a hyperparameter to tune. However, a default decision threshold of 0.5 was used for this work because it yielded good performances for the model. The advantages of this architecture are the clear interpretability of the probability of a flight experiencing a given event or not and the easier learning task that the model has to accomplish.

b. Multiple Output Node. The binary architecture can also be changed to include multiple output nodes, such that the output after the last max pool layer is of size (*batchsize*, *c*) where *c* is the number of classes to predict. The sigmoid layer before the final max-pooling ensured each class/anomaly; the probability of experiencing any of

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Table 3 Hyperparameters and search space

Hyperparameter	Search range
Convolutional layers 1, 2, 3 kernel sizes	[8, 5, 3], [6, 3, 2]
Convolutional layers 1, 2, 3 output channels	[10, 15, 20], [16, 32, 64], [32, 64, 128]
Learning rate	0.001, 0.0001
Weight decay (L2 regularization)	0.01, 0.001, 0.0001
Mini-batch size percent	1% of training set size
Optimizer	ADAM [38]
Number of epochs	30

them is bounded between 0 and 1. Indeed this allows for multi-labeling, which is a use case in aviation because multiple events could occur during the same flight. Using a sigmoid layer is different from using a softmax layer, which would require the sum of the probabilities of each event to be 1. This architecture was preferred over using a softmax layer because it could handle both multiclass and multilabel problems. In fact, for this work, the model was trained using flights that experienced only one anomaly at a time and therefore learned to keep the probabilities of nonoccurring events low while keeping the ones of occurring events higher. Thus, for each flight, the class with the highest probability was chosen as the final output class. The advantage of this architecture is the convenience of training one model instead of multiple binary classifiers.

3. *Hyperparameter Search*

The chosen architecture is novel for precursor mining. Thus there is a lack of knowledge in choosing the model's parameters and how to train it. A hyperparameter search was performed to determine the optimal parameters of the architecture for each event to mitigate this problem. A grid search is the most straightforward strategy to perform a hyperparameter search. It entails searching through the space created by all possible combinations of hyperparameters. In practice, each parameter is given an array of values to try. The model is then evaluated using metrics described in Sec. III.B.4 or each of the parameter combinations. Table 3 defines the hyperparameter space containing 36 possible combinations.

The data set was divided into three subsets: the training set, the validation set, and the testing set, as seen in Fig. 8. Stratified subsets are used to ensure that the proportion of nominal flights to abnormal flights remain consistent in each subset. Additionally, a stratified mini-batch was used during the training process. The training set was used to train the model using a given hyperparameter combination. Google Colab's Tesla V100 16 GB of RAM GPU was leveraged to train the models. The performances of the models on the validation set were used to determine the best model.

4. *Classification Metrics*

Several metrics are used to evaluate the model's performance quantitatively. Evaluating the best model from the hyperparameter search on new unseen data (i.e., on the test set) allows for its selection. The best model correctly forecasts the occurrence of an adverse event and is best suited to determine the aircraft's parameters that are the most correlated to the event. The following metrics were used for the model selection and evaluation:

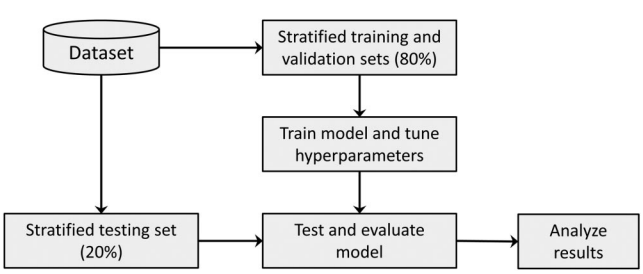

Fig. 8 Data split for model training, hyperparameter tuning, and model evaluation.

Table 4 Confusion matrix

	Predicted: no event	Predicted: event
Actual: no event	True negative (TN)	False positive (FP)
Actual: event	False negative (FN)	True positive (TP)

1) *Confusion matrix:* It contains the counts for true positives (TPs), true negatives (TNs), false positives (FPs), and false negatives (FNs) [39], as shown in Table 4.

2) *Precision:* It measures the proportion of positively labeled examples that are positive [40] (the closer to one, the better). The following equation defines the precision:

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2}$$

3) *Recall:* It measures the fraction of positives labels that were detected [40] (the closer to one, the better). The following equation defines the recall:

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$

4) *F1 score:* The harmonic mean of precision and recall [40] (the closer to one, the better). The following equation defines the F1 score:

$$\text{F1score} = \frac{2 \times \text{TP}}{2 \times \text{TP} + \text{FN} + \text{FP}} \tag{4}$$

5) *Distance from ADOPT (DFA):* A metric created to measure the resemblance in the precursor score ranking of IM-DoPE and ADOPT [26]. Assuming that precursor scores p_i for feature i ranked between 0 and 1, N flights, and d feature, the DFA for each combination of hyperparameters is given by

$$sse_j = \frac{1}{d}\sum_{i}^{d}(p_{i,\text{IMDoPE}} - p_{i,\text{ADOPT}})^2 \tag{5}$$

$$\text{DFA} = \frac{1}{N}\sum_{j}^{N} sse_j \tag{6}$$

DFA for a combination of hyperparameters is the mean sum square error between the precursor scores of ADOPT and IM-DoPE. The closer to zero, the more alike the rankings of the two algorithms are. The sum square errors of the precursor score are averaged across all features [Eq. (5)] and all flights [Eq. (6)]. Experts have validated ADOPT for few events such as a high-speed exceedance and can thus be used as a second-hand validation if no human expertise is available.

C. Precursor Identification

1. *Raw Precursor Scores*

Once the model achieves acceptable performances, it can be used to identify precursors. The precursor score p_i, t of each feature i at time step t is extracted directly from the architecture. As seen in Fig. 9, the raw precursor score for each feature can be extracted from the last convolution layer of each head. The raw scores are bounded between 0 and 1 due to the sigmoid function used in the fourth convolutional layer. From empirical results, it was observed that the model learns to set nonimportant parameters to 0.5. As an example in Fig. 9, the rudder position, the rudder pedal, and the spoiler position are deemed unimportant by the model because their individual scores remain at 0.5 for the last 20 n miles before a 1000 ft. On the other hand, the radio altitude was flagged as a precursor. As the less important features go through the different convolutional layers, they get reduced to zero. This zeroed-out feature is then passed through the sigmoid activation resulting in a raw score of 0.5.

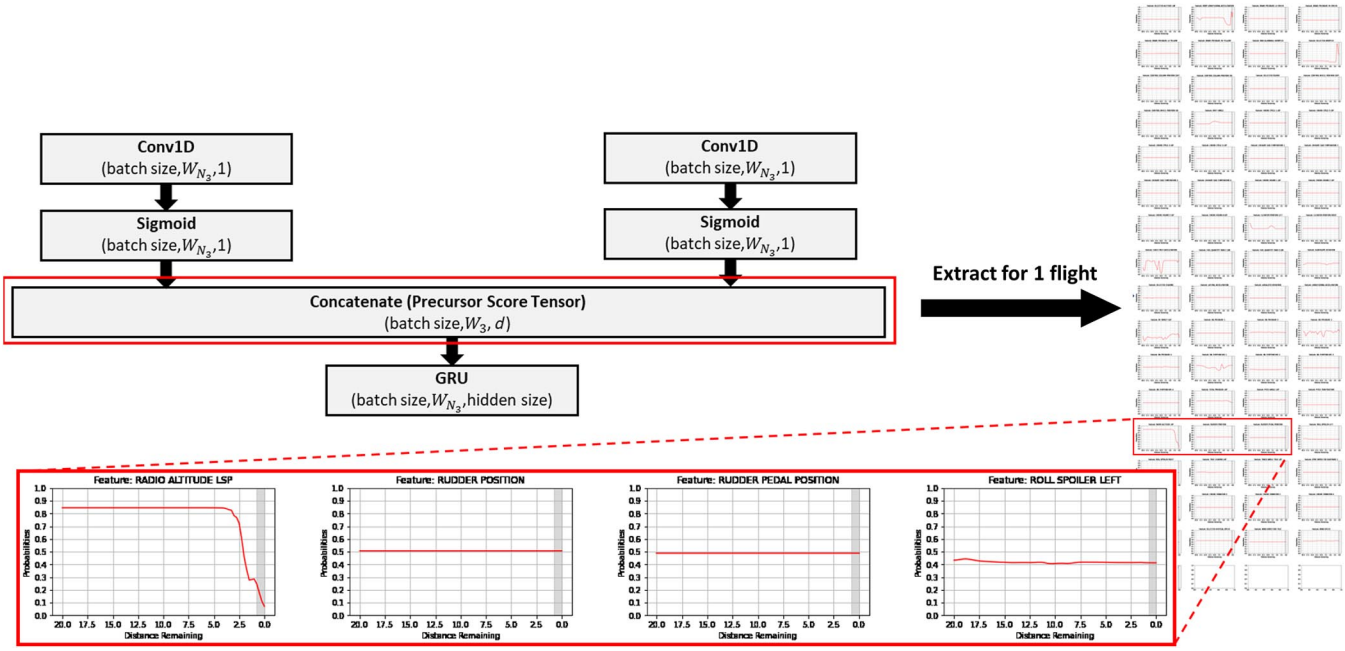

Fig. 9 Example of the extraction of precursors for a high-speed event.

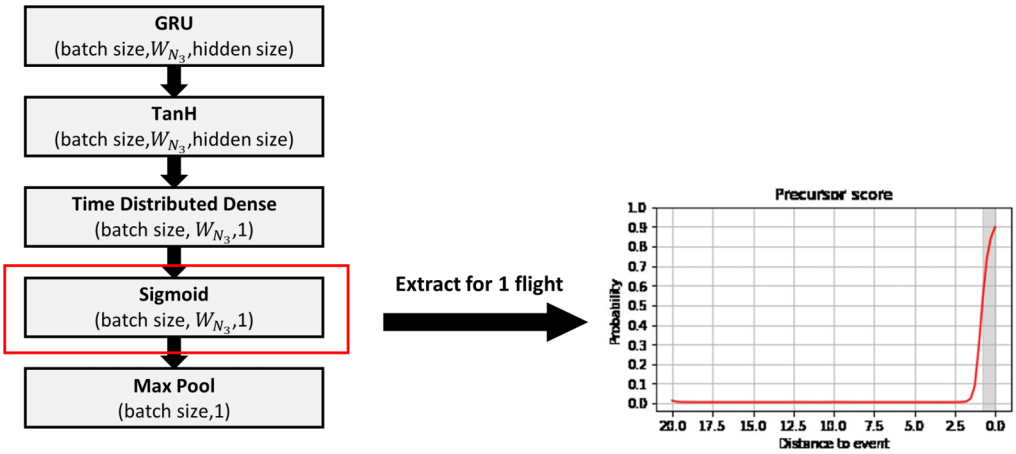

Fig. 10 Extraction of precursors score over time from dense layer for a high-speed event.

The identified precursors have an instance of time (precursor score over time) at which they are more correlated to the event of interest (grayed area in Figs. 9 and 10). The combination of GRU and dense layer allows for the extraction of these time instances, as seen in Fig. 10. Similar to ADOPT, abnormal flights will have the precursor score increase toward 1 and nominal flights will fall toward 0. Additionally, because the precursors are more correlated to an event at time windows identified by the GRU, an overall score can be given to the precursor by averaging its $p_{i,t}$ values within an identified window T.

2. *Adjusted Precursor Scores*

A more natural way to express the precursor score is to have a score of zero for parameters that are not important. Therefore, the raw precursor scores were adjusted. The following equation, therefore, defines the adjusted precursor score p_i of feature i:

$$p_i = \frac{1}{|T|}\sum_t (|p_{i,t} - 0.5|), \quad \forall\, t \in T \quad \text{where } T \subseteq L \tag{7}$$

where $|T|$ is the number of time steps within the time window T. The adjusted precursor score is thus bounded between 0 and 0.5. From this definition, it can be seen that features with higher scores are flagged as precursors.

IV. Results and Discussion

A. Quantitative Results

1. *Multiple Binary Classifiers*

For each event, 36 combinations of hyperparameters were evaluated, and the best classifiers were selected. Table 5 summarizes the quantitative results obtained when evaluating the models on the test set.

While the best MHCNN-RNN model achieves high performances, better results for the classical classification metrics are obtained for the high-speed event due to the higher number of training examples and the smaller imbalance in the classes. However, the larger DFA metric for the high-speed event suggests a more significant difference with ADOPT feature ranking. Note that the DFA values obtained are

Table 5 Model evaluation results (binary classifications with multiple binary classifiers)

Event	Algorithm	F1 score	Precision	Recall	DFA
High speed	IM-DoPE	0.90	0.84	0.97	0.0392
High path angle	IM-DoPE	0.83	0.81	0.87	0.0152
High speed	ADOPT	0.88	0.90	0.86	N/A
High path angle	ADOPT	0.70	0.56	0.90	N/A

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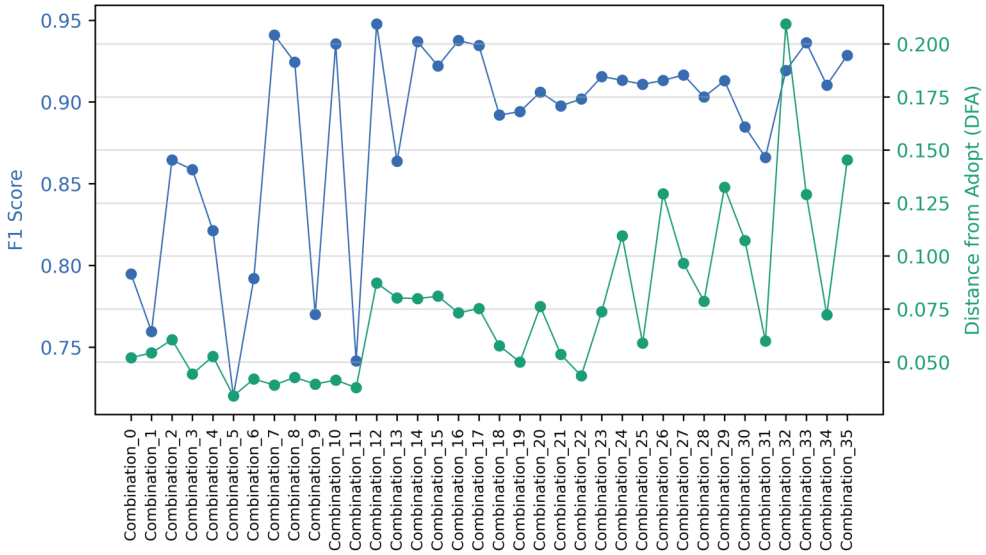

Fig. 11 F1 score and DFA for all combinations (high-speed event).

still relatively good as it is among the lowest DFA values obtained as seen in Fig. 11.

Additionally, the best model of IM-DoPE performs better than ADOPT when using the same training set and testing set. However, it is important to note that while a grid search was performed for IM-DoPE for each event, the default setting of ADOPT was used. Some models obtained during the grid search have lower scores than the ones obtained from using ADOPT default settings. For example, in Fig. 11, several models reached scores lower than 0.88.

The binary classifiers are then combined as explained in Sec. III.B.2 in order to obtain a multiclass classifier. Similar to binary classification, each metric can be computed for each class. For instance, the precision of the high-speed class is the number of flights that faced a high-speed event over the total number of flights that were labeled as a high-speed event by the classifier. Table 6 presents similar scores as in Table 5, showing that each model learns to identify its positive class correctly even when combined. The confusion matrix in Table 7 confirms this behavior, as the numbers of FP and TN for each class are relatively low.

2. *Multiple Output Nodes*

Similarly, the multiple output nodes architecture was trained using the same list of hyperparameters. After training, the best set of hyperparameters can be chosen using the validation set, and the final performances of the model on the unseen test set can be evaluated. The results of this evaluation are presented in Table 8. On the one hand, from the results obtained, it can be seen that there are slight improvements in the F1 score of the nominal and high-speed classes thanks to the respective increase in the recall and precision scores. On the other hand, the ability to accurately predict the high-path-angle anomalies decreased. This model architecture can find most flights that experienced the high-path-angle event, but has a higher number of FPs than the other multiclass model leading to a much lower F1 score.

Table 6 Model evaluation results (multiclass classifications with multiple binary classifiers)

Event	F1 score	Precision	Recall
Nominal	0.91	0.97	0.85
High speed	0.90	0.84	0.97
High path angle	0.82	0.80	0.83

Table 7 Confusion matrix (multiclass classifications with multiple binary classifiers)

	Predicted nominal	Predicted high speed	Predicted high path angle
Actual nominal	1717 (47%)	263 (7.2%)	33 (0.9%)
Actual high speed	38 (1.0%)	1422 (39%)	1 (2.7×10^{-4}%)
Actual high path angle	16 (0.44%)	12 (0.33%)	138 (3.8%)

B. Identified Precursors

Even though both the multiple binary models and the multiple output nodes model can be used along with the precursor identification part of the IM-DoPE methodology, the combination of multiple binary classifiers yielded better results. It, therefore, was chosen to perform the identification of precursors. The results are presented in the following subsections.

1. *Average Adjusted Precursor Scores*

As previously mentioned, the final model is expected to predict adverse events and identify their precursors. The adjusted precursor score can be obtained for each feature and each flight. The average adjusted precursor score can be determined when only considering true positive flights. The precursors can be identified on a fleet level, as seen in Table 9 for the top 5 precursors of each event. Noticeable differences are observed in the precursor rankings for the two events. For example, the glideslope deviation and the pitch angle are characteristics of a high-path-angle event, whereas the N1 target relates to engine power that can be related to speed. Some resemblances are also observed; for instance, the altitude is seen as the precursor for both events. This is expected because the altitude above touchdown is used to define both events. The trained model can also be used to analyze individual flights. Two flights experiencing a high-speed event and a high-path-angle event are analyzed in Secs. IV.B.2 and IV.B.3, respectively.

Table 8 Model evaluation results (multiclass classifications with multiple output nodes)

Event	F1 score	Precision	Recall
Nominal	0.93	0.93	0.92
High speed	0.92	0.95	0.89
High path angle	0.69	0.55	0.92

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Table 9 Average adjusted precursor scores for high-speed and high-path-angle events

Precursor rank	Average adjusted precursor score: High-speed event	Average adjusted precursor score: High-path-angle event
1	Altitude: 0.31	Radio altitude: 0.31
2	Radio altitude: 0.28	Glideslope deviation: 0.24
3	N1 target: 0.25	Pitch angle: 0.21
4	Body longitudinal acceleration: 0.23	Altitude: 0.18
5	Lateral acceleration: 0.14	Flight path acceleration: 0.12

2. High-Speed Event Flight Analysis

When performing inference using the trained model, the precursor scores can be extracted from the model's MHCNN outputs. Following previously highlighted steps in Sec. II.C.2, the adjusted precursor score can be computed and used to identify the precursors for a flight. Figure 12 shows the precursor ranking obtained for a flight that experienced a high-speed event. ADOPT identified similar precursors for this flight, and the top 5 precursors were the altitude, the radio altitude, the flight path acceleration, the N1 target, and the total pressure. Once the precursors are discovered, the aircraft's parameters can then be plotted to assess them.

In particular, the time series is of interest when the precursor score over time is greater than the 0.5 threshold, which occurs at around 1.5 n miles away from a 1000 ft above touchdown in Fig. 13 (gray-shaded area). In the figure, the red and blue curves in the first cell represent the precursor at different time steps obtained from the high-speed and high-path-angle classifiers, respectively. Because the flying conditions do not correspond to a high-path-angle event, the classifier for the event cannot find precursors. In contrast, the classifier for the high-speed event detects abnormal patterns leading to the identification of precursors. Note that the model would have confidently classified the flight as normal if only the high-path-angle binary classifier was used. In the remaining cells, the dotted blue line represents the feature value. For instance, this flight reached a pitch angle of 5 deg 7.5 n miles away from 1000 ft. The dotted black line represents the mean nominal values for each feature at every time step, and the shaded purple region is defined as ± 2 standard deviations away from the mean. A potential cause of the excessive speed experienced by the flight at 1000 ft can be inferred from the figure. The body longitudinal time series shows an initially higher than nominal positive acceleration followed by a negative one, suggesting that the pilots attempted to reduce speed. In addition, the lower than nominal flight path acceleration and lower than nominal pitch angle in the last 2.5 n miles warrant more investigation into the causes of the event.

3. High-Path-Angle Event Flight Analysis

Similar to the high-speed event case, the precursors to the high-path-angle event can be probed from the model. Again, the identified precursors provide a list of aircraft parameters that can be assessed through visualization. For the selected flight, the presence of a dominating precursor is observed. Indeed, there is a larger difference between the top two precursors for the high-speed event, as shown in Fig. 14. This larger difference suggests a highly abnormal glideslope deviation. Moreover, the glideslope was also identified by ADOPT as the top parameter. Other important parameters were the pitch angle, the radio altitude, the airbrake position, and the flight path acceleration.

The abnormal behavior is confirmed by Fig. 14 because the glideslope deviation is much greater than two times the standard deviation. The potential cause of the event is likely mainly related to the high deviation in the glideslope. However, other precursors such as the pitch angle and the flight path acceleration are also identified and observed to have abnormal patterns. It is important to note that these two parameters can also be precursors to the high-speed event, as previously observed. This flight was not classified as a high-speed event but some of the parameters behaved similarly to how they would behave during such an event. This led the high-speed event classifier to increase the precursor probability over time to be closer to 1. Ultimately, the multiclass classifier correctly labeled this flight as a high-path-angle event because the high-path-angle classifier was more confident than its high-speed counterpart.

C. Discussion of Results

The MHCNN-RNN architecture yields satisfying quantitative results. The scores obtained from the conventional classification metrics showed that the model is capable of extracting information from the data and use it to accurately determine whether a flight will experience an adverse event or not. The precursor identification step aims to discover the parameters that help the model make correct

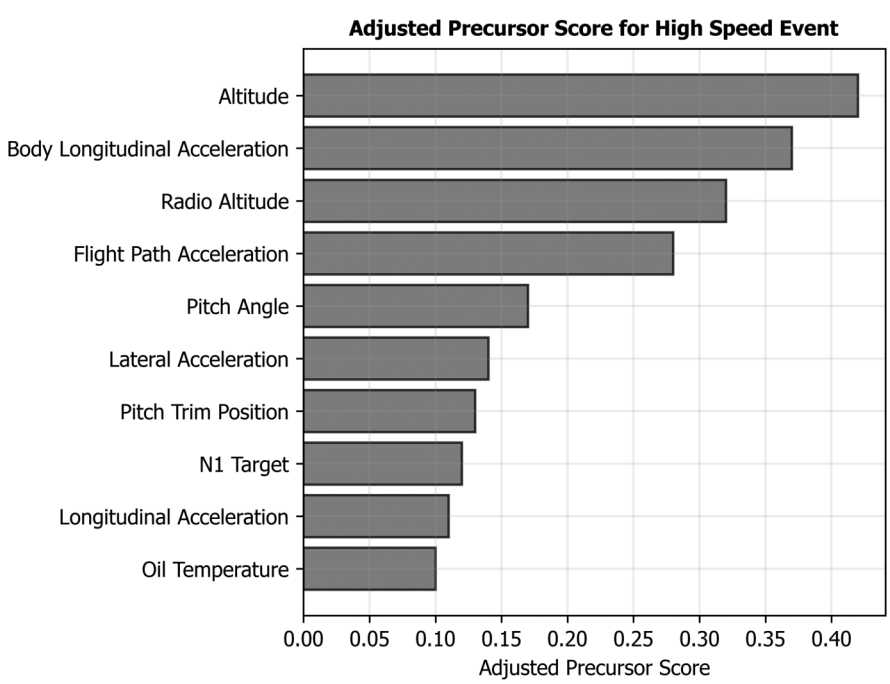

Fig. 12 Precursor ranking for a high-speed event.

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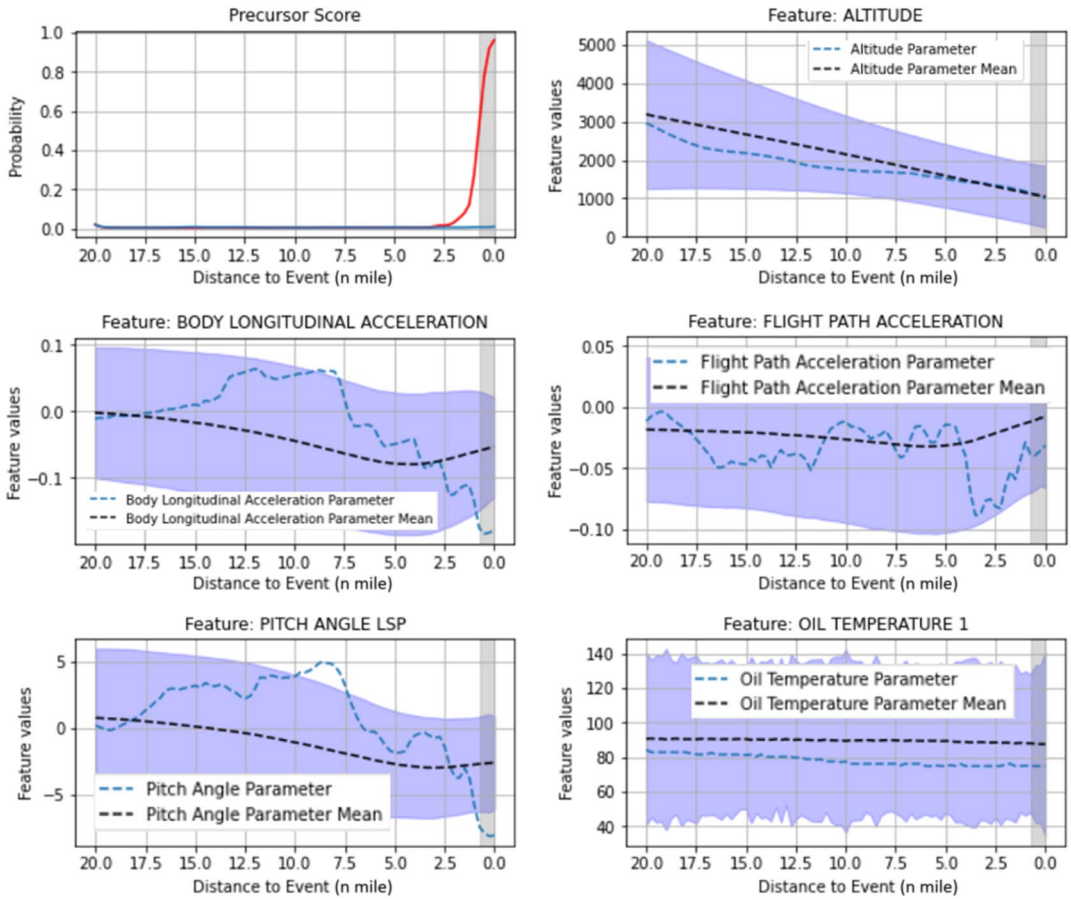

Fig. 13 Precursor score and aircraft's parameters during a high-speed event.

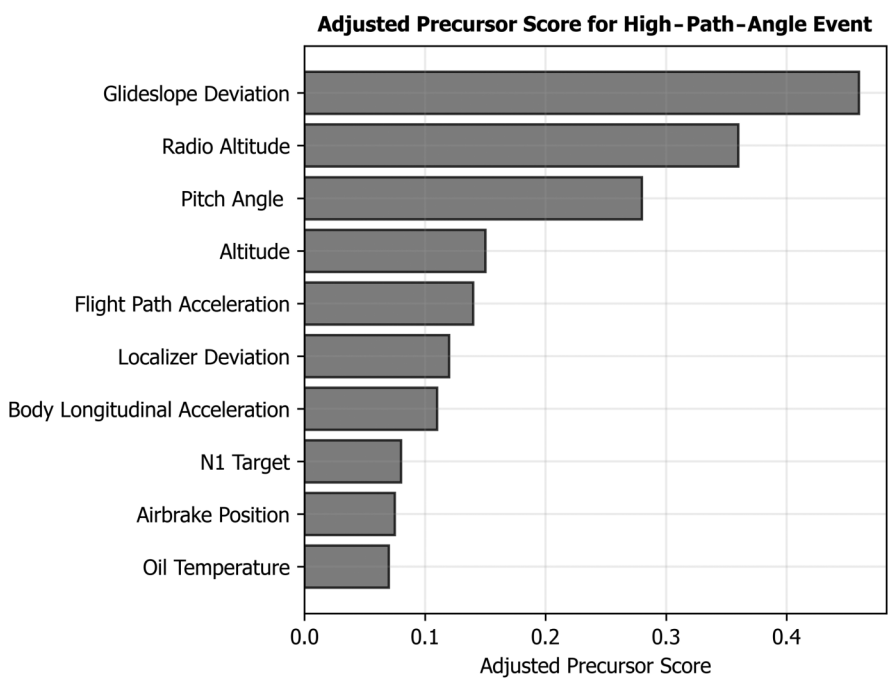

Fig. 14 Precursor ranking for a high-path-angle event.

predictions. The methodology developed to compute the adjusted precursor score essentially measures the absolute deviation from the reference values of 0.5 in the feature map of the last CNN of each head. Although the theoretical interpretation is still an area of research, multiple experiments showed a correlation between the parameters with the highest adjusted precursor scores and the events studied. It was observed that different precursors are discovered for different events, which is expected. High-speed top precursors relate more to poor energy management (kinetic and potential), whereas the precursors for the high path angle include trajectory-related parameters, as seen in Table 9. The identified precursors are partially validated given their resemblances with ADOPT-identified

precursors, characterized by low DFA values. Additional support toward the validation of the discovered precursors is obtained through visualization and observing that the precursors exhibited abnormal behaviors outside of nominal operations.

Some unexpected behavior was also observed from the model. Because some parameters deviate from nominal operations, they are flagged as precursors. For example, the oil temperature for the high-speed event is identified, though it might not provide insights into potential causes of excessive speed. That is a limitation because, from flight to flight, some of the parameters flagged might not be relevant. There is currently no strategy to determine when the irrelevance starts, given a list of precursors. A human expert is thus required to follow up on the identified precursors to assess their importance.

In practice, the offline deployment of the precursor model could allow FOQA analysts to automatically flag flights experiencing predefined anomalies without any human asking initial queries and highlight parameters that could be related to it. The analysts would then be required to review the flagged flights and parameters, but their work would have been alleviated as the precursor model would provide a head start on where to look. New patterns in operations could be discovered using this data mining technique if the model identifies previously unknown adverse trends. The knowledge of these trends could then be used to update the pilot's training to avoid unwanted behaviors in the future. In addition, the precursor model's outputs (identified precursor and region of time of precursor activities) could be used to develop rule-based methods to label precursors, for which there is currently a lack of labeled data. For example, from Fig. 13, analysts could define a high-speed event precursor to be active when the pitch angle and body longitudinal acceleration are lower than −5 deg and −0.1, respectively, when the aircraft is 1.5 n miles away from 1000 ft. On the other end, the online deployment of such a model would warn pilots of a predicted adverse event, allowing them to take corrective actions ahead of time. For example, in Figs. 13 and 15, pilots would have been able to take action a few nautical miles before reaching the 1000 ft mark.

Finally, the work proposed in this paper advances the state-of-the-art precursor mining model by completely getting rid of the post-processing step required to identify precursors. Removing this step saves computing power and time and removes the bias of a sensitivity analysis, which potentially misses interactions of parameters. In addition, the methodology proposed and compared two ways of tackling multiclass classification, which was lacking in the precursor mining literature. The results showed that developing a model composed of multiple binary classifiers yields a better model with higher F1 scores. Although the performance of multiple binary classifiers is better than a classifier with multiple output nodes, the end-user would have to weigh in the added complexity in deploying and maintaining that many models.

V. Conclusions

The work presented in this paper tackles the precursor mining task. A public flight data set was leveraged to create FOQA-like labels. After performing the required preprocessing steps, a novel architecture for the task was developed to take advantage of the multi-instance learning framework, the feature extraction capabilities of CNNs, and RNNs' temporal pattern recognition capabilities. On the one hand, binary classifiers were trained to predict high-speed and high-path-angle events, and on the other hand, the MHCNN-RNN architecture was modified to handle multiple outputs. In both cases, a grid search was implemented to determine the best parameters for the neural networks and thus the best model for predicting the safety events. The best two models were then combined to form a unique multiclass classifier for the multiple binary classifier case. For both multiclass extensions, the final models were evaluated on a test set.

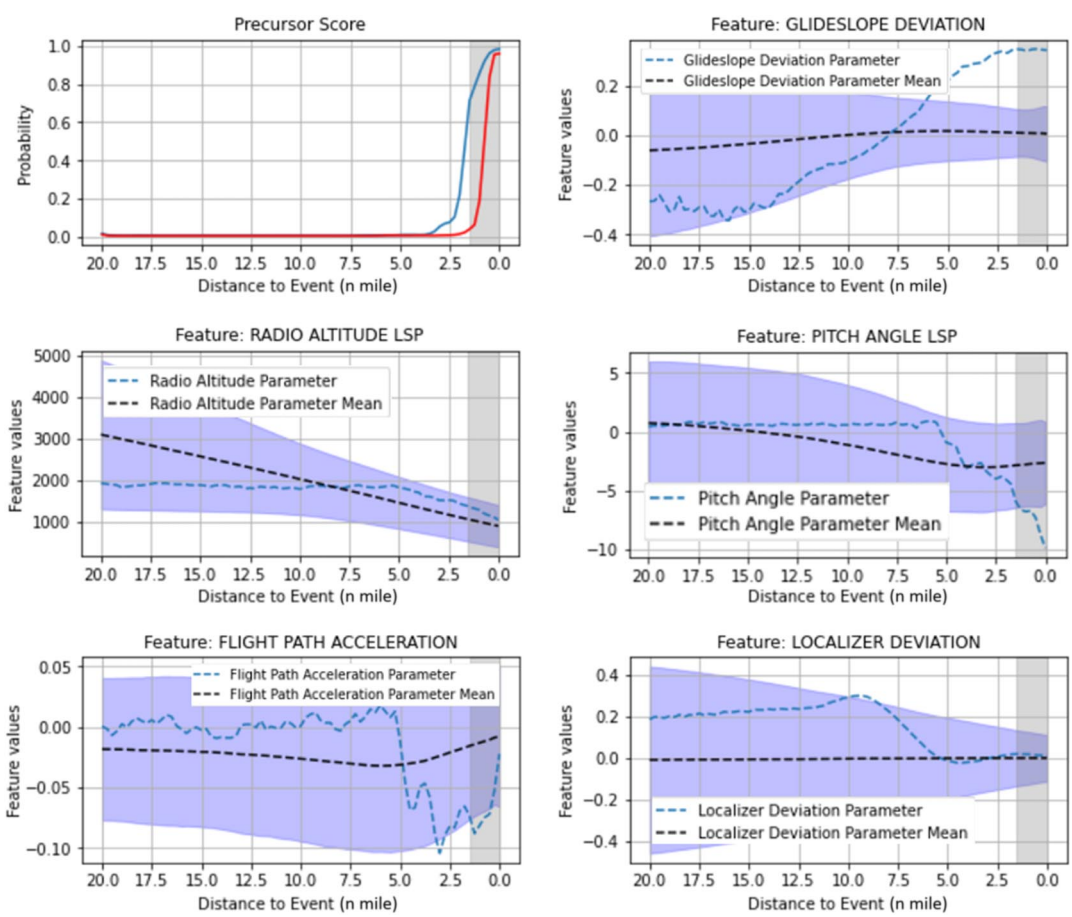

Fig. 15 Precursor score and aircraft's parameters during a high-path-angle event.

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High scores were observed for classification metrics such as F1 score, precision, and recall, in particular when combining binary classifiers.

Furthermore, the binary models were then used to identify precursors and provide the average precursor score across all flights that experienced each of the two events. Finally, visualizations were used to observe the behaviors of the identified precursors, which exhibited patterns different from normal flight operations. However, the algorithm tends to flag unwanted precursors, and human experts must validate the identified precursors. Future work will include enhancing the interpretability of the precursor score tensor. Current empirical results show that the parameters that deviate from 0.5 are understood to be correlated to the event of interest. The meaning of the direction of the deviation (greater than or lower than 0.5) needs to be investigated. Further improvements to the model can be made toward increasing the prediction window and allowing the classification of unknown precursors instead of defaulting unknown behaviors to nominal. Additionally, the lower scores for the modified MHCNN-RNN could be due to the class imbalance, especially the lower number of high-path-angle events. Future work will also explore methodologies to handle class imbalance and extend the learning task to a multilabel problem.

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